MILITARY SPECIFICATION

SLINGS, CARGO, PAULIN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers heavy duck body, rope and nylon webbin reinforced slings.

1.2 Classification. Slings shall be the following types as specified (see 6.2):

Type I - For loading bagged cargo.
Type II - For lifting debris.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3940

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SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
T-R-605	- Rope, Manila and Sisal.
T-T-616	- Treatment: Mildew Resistant, for Rope and Cord.
T-T-891	- Twine, Linen.
V-T-285	- Thread, Polyester.
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-C-428	- Cloth, Duck, Cotton; Fire, Water, Weather, and Mildew Resistant.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

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MIL-T-3530	- Thread and Twine; Mildew Resistant or Water Repellant Treated.
MIL-W-23223	- Webbing, Nylon, Slotted.

STANDARDS

FEDERAL

FED-STD-751	- Stitches, Seams, and Stitching.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The slings shall consist of a cotton duck body reinforced with manila rope for type I and with nylon webbing for type II.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be new and unused and subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control (MADPAC). The slings shall be fabricated from compatible materials, suitably finished, to provide protection from corrosion and other forms of material deterioration. Assurance shall be given in the form of a finish plan which clearly identifies the specific finish or treatment to be used on the various components and assemblies, to provide protection against the various forms of material deterioration in the applicable operating and storage environments to which they may be exposed.

3.3.2 Cotton duck. Cotton duck shall conform to CCC-C-428, type I, class 1. The basic cloth shall conform to CCC-C-419, type I. The weight of cotton duck shall be commercial No. 6.

3.3.3 Rope, manila. The rope shall be three stranded standard lay manila conforming to T-R-605, type M, class 2, and shall be treated for mildew resistance in accordance with T-T-616.

3.3.4 Nylon webbing. The nylon webbing shall conform to MIL-W-23223, class B, type III for loop straps and type X for the border material. It shall be in olive drab color.

3.3.5 Polyester thread. The polyester thread shall conform to V-T-285, type I, class 1, size 3, natural color.

3.3.6 Twine. Twine shall conform to T-T-891, type II, 5-ply and shall be treated for mildew resistance in accordance with MIL-T-3530, type I, class 1.

3.4 Construction.

3.4.1 Type I.

3.4.1.1 Type I sling. The type I sling shall consist of a one-piece duck body reinforced with manila rope as shown in figure 1, and as specified herein. Hems shall be not less than 2-1/2 inches wide.

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3.4.1.2 Stitches, stitching, and seams. Stitches, stitching, and seams shall be in accordance with FED. STD. No. 751. Stitching shall conform to stitch type 301. Where selvaged edges are hemmed, stitching type EFa-2 shall be used; where cut edges are hemmed, stitching type EFb-2 shall be used. The turn-under in stitching type EFb-2 shall be not less than 1 inch. Stitches shall be 1/2 inch plus or minus 1/16 inch from the inner edge of the hem. Machine stitching shall be not less than 6 stitches per inch and shall be back-stitched not less than 1 inch at each break. All stitching shall be back-stitched at the ends to prevent raveling, except where ends are turned under in a hem or held down by other stitching. Thread tension shall be maintained to prevent loose stitching and to insure the embedding of the lock in the center of the materials used. All rope shall be stitched to the duck at each point of contact with each strand and as shown in figure 1. Each stitch shall pass completely around and under one strand of rope. Surplus duck may be allowed between each stitch to transmit all stress to the rope when sling is in use. Hand stitching shall be done with six parts of waxed, twisted, and rewaxed twine.

3.4.1.3 Reinforcing ropes. Reinforcing ropes (3-3/4 inch circumference, 1-1/4 inch diameter) shall be unlaid to the four points of intersection. At points of intersection, the strands of the one intersecting rope shall be passed through the strands of the other. Ropes shall be relaid and the ends spliced together with a short splice.

3.4.1.4 Framing. The outside framing rope (1-1/2 inch circumference, 1/2-inch diameter) shall extend the entire perimeter of the sling proper. Framing rope shall pass between the strands of diagonal ropes at points of intersection.

3.4.1.5 Splices. All splices forming an eye shall be four-tuck eye splices. All other splices shall be short splices, having three tucks each way. End of strands shall be whipped with twine in two places: 2 inches and 3 inches from the splice and clipped 1/2 inch from end whipping.

3.4.2 Type II.

3.4.2.1 Type II, for lifting debris. The type II sling webbing shall consist of strips of cotton duck reinforced with nylon webbing as shown in figures 2 and 3 and as specified herein. Each strip shall be of sufficient length to provide the finished 12 foot dimension without piecing. Strips shall be not less than 20 inches wide before sewing, nor shall the width of any strip vary more than 1/4 inch along its entire length. Corners shall be turned back and stitched to the duck body. Hems and seams shall be not less than 2-1/2 inches wide. Nylon webbing shall be doubled at beckets or loops and extended on duck canvas as shown in figure 3.

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3.4.2.2 Stitches, stitching, and seams. Stitches stitching, and seams shall be as specified in 3.4.1.2, except all nylon webbing shall be stitched to the duck as shown in figures 2 and 3, with zigzag reinforcing stitch where indicated.

3.5 Lifting capacities. The slings shall lift a load of 2700 pounds without elongation, deformation, or rupture of any part.

3.6 Identification marking. The sling shall be identified in accordance with MIL-STD-130. Marking shall be legibly printed or stenciled with black waterproof ink or stencil paint on the cotton duck body.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsbility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and teste in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article sling shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article sling shall be tested as specified in 4.5.2.1. Failure of this test shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. AQL shall be 1.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection of the entire log represented by the sample.

4.5 Inspection procedure.

4.5.1 Examination. The sling shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Finish plan not adequate.
103. Dimensions not as specified.
104. Hems less than 2-1/2 inches wide.
105. Corners not turned back and stitched to the duck body.
106. Machine stitching not as specified.
107. Reinforcing ropes not as specified.
108. Splices not as specified.
109. Framing ropes not of specified circumference.
110. Nonconformance to dimensions shown in figures.
111. Nylon webbing not as specified.

Minor

201. Ravelled threads and loose stitching.
202. Identification marking illegible, incorrect, incomplete, or missing.

4.5.2 Test.

4.5.2.1 Proof load. A 2700 pound uniform load shall be placed on the reinforcing ropes of the sling. A vertical hoisting force shall be applied to the becket or loops of the reinforcing ropes, raising the sling assembly with proof load until free from any support. The loaded sling assembly shall be raised and lowered three times. Each time it shall be held suspended for an interval of 30 seconds. Nonconformance to 3.5 shall constitute failure of this test.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Lot. A lot shall consist of all completed packs offered for examination at one time.

4.6.1.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.4 Examination. Samples selected in accordance with 4.6.1.3 shall be examined for the following defects. AQL shall be 2.5 percent defective.

No.	Defect	A	Comm.
112.	Bundle not compact as specified.	5.1.1	5.1.2
113.	Rope extensions not inside bundle as specified.	5.1.1	5.1.2
114.	Bundle not wrapped in waterproof material as specified.	5.1.1	
115.	Shipping container not as specified.	5.2.1	5.2.2
116.	Strapping not as specified.	5.2.1	
117.	Marking not as specified.	5.3.1	5.3.2

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each sling shall be folded or rolled into a compact bundle with the rope extensions inside the bundle. The bundle shall then be secured with cotton tape or twine. Each bundle shall be further protected by one of the following methods:

a. Wrapped with barrier material conforming to PPP-B-1055, class El and each fold, lap and seam sealed with tape conforming to PPP-T-60 type IV, class 1, or

b. Wrapped with polyethylene material conforming to L-P-378, type II, class 1, grade optional, finish 1, film thickness 0.0060 inch and each fold, lap and seam sealed by a heat seal or by the tape specified herein.

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5.1.2 Commercial. Each sling shall be folded or rolled into a compact bundle with the rope extensions inside the bundle. The bundle shall then be secured with tape or twine. Additional preservation shall be as specified in MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. The slings, preserved as specified in 5.1, shall be packed in close fitting boxes conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional, in quantities not to exceed the weight limitations of the applicable box specification. Box closure and strapping shall be as specified therein except that strapping shall be flat and finish shall be B. Skid application shall be as specified herein.

5.2.2 Level B. Level B packing shall be as specified for level A in 5.2.1 except that the boxes shall be domestic type or class and that strapping need not be finish B.

5.2.3 Commercial. The slings, preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Level A. In addition to any special or other identification marking required by the contract or purchase order (see 6.2), each shipping container shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be as specified in MIL-STD-1188.

6. NOTES

6.1 Intended use. The type I sling is intended for use in loading of bagged cargo such as flour and beans. The type II sling is intended for use in lifting debris from the hold of vessels.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of sling required (see 1.2).

c. When a first article is required for inspection and approval and the number of units required (see 3.2).

d. Degree of preservation and packing required (see 5.1 and 5.2).

e. Any special markings (see 5.3.1).

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6.3 First article. When a first article inspection is required, the item shall be tested and should be a first production model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

Custodians:
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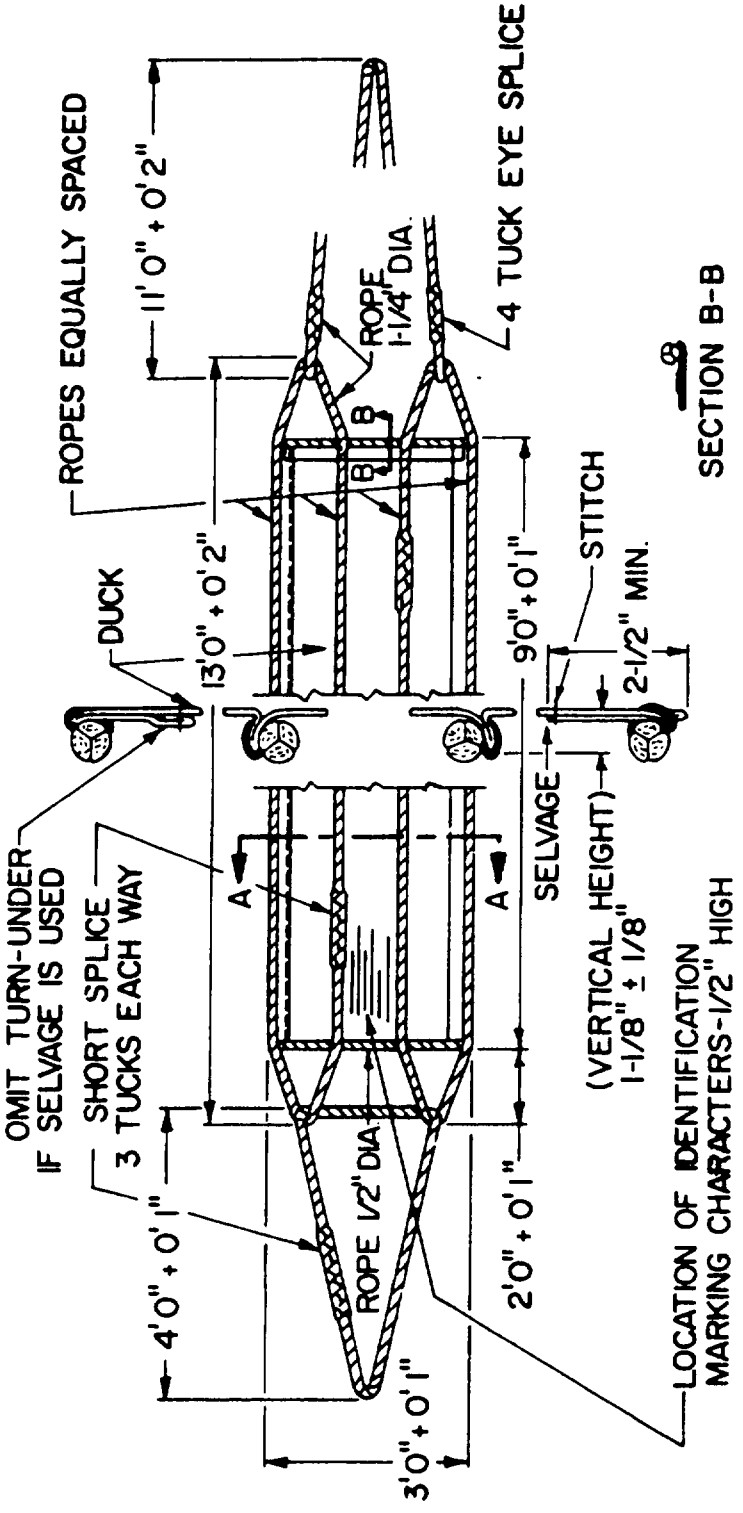
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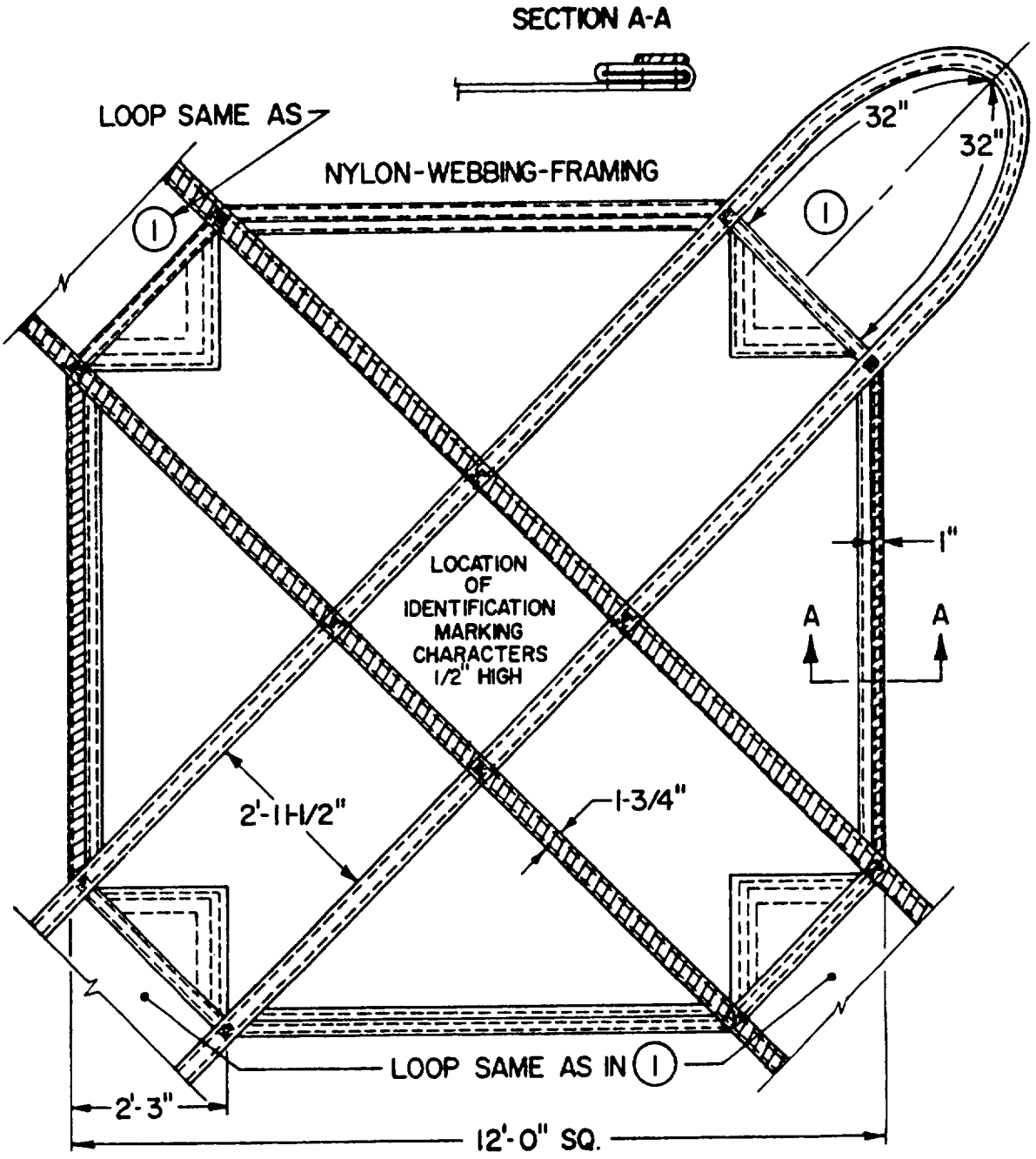
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FIGURE 1. Sling, cargo, paulin, type I.

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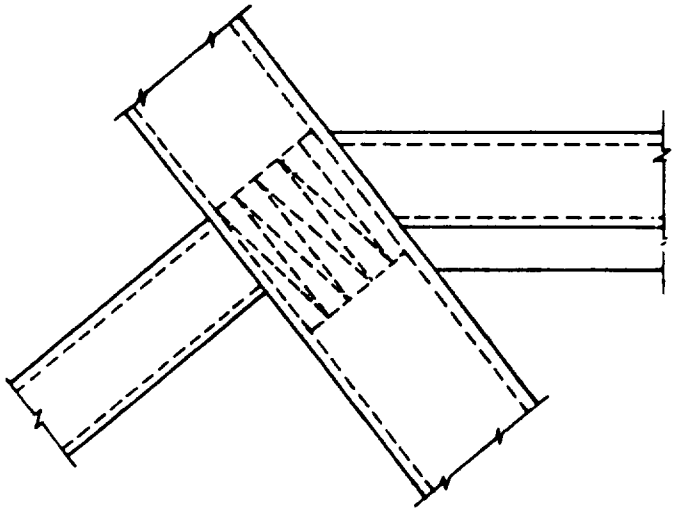


TOLERANCE OF -2 INCHES ON ALL OVERALL DIMENSIONS

FIGURE 2. Sling, cargo, paulin type II.

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ZIG ZAG REINFORCEMENT STITCH

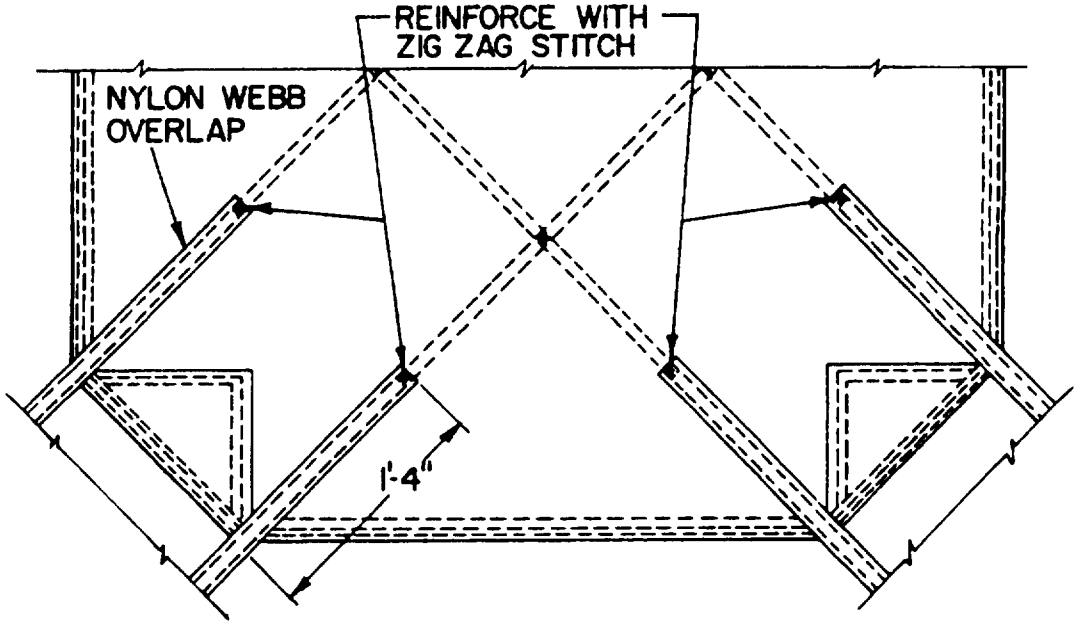


FIGURE 3. Backside-sling.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-3112C	Slings, Cargo, Paulin

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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